

The correspondence lines pair each strip of the upper figure with its mate below: p_1 with q_1 , p_2 with q_2 , p_3 with q_3 . By hypothesis every such pair shares one ultimate ratio k , so that $p_i/q_i \rightarrow k$ each to each respectively. Then, by composition, the sum of all the upper parallelograms is to the sum of all the lower parallelograms in that same ratio k , since adding terms of equal ratio preserves the ratio. Finally, by Lemma III each rank of parallelograms is ultimately equal to its own curvilinear figure — so the sums may be replaced by the whole figures, giving $AacE : PprT = k$. Thus two figures whose corresponding strips share an ultimate ratio are themselves in that very ratio. *Q.E.D.*